

Quantum impact on symmetric cryptography

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New quantum-resistance security levels



- “bits of security” does not correlate with practical security when attacks cannot be effectively parallelized. A CRQC that can break RSA in a few hours would not pose any practical threat at all to symmetrical algorithms such as AES-128 and SHA-256. **Even with millions of CRQCs it would take millions of years.**
- 128-bit algorithms will remain secure for the foreseeable future and NIST new quantum-resistance security levels are defined using the quantum resistance of symmetric-algorithms.

Table 2. Security strength categories for post-quantum algorithms

Security Category	Minimum Attack Cost	Example
1	Same as key search on a block cipher with a 128-bit key	AES-128
2	Same as collision search on a 256-bit hash function	SHA-256
3	Same as key search on a block cipher with a 192-bit key	AES-192
4	Same as collision search on a 384-bit hash function	SHA3-384
5	Same as key search on a block cipher with a 256-bit key	AES-256

No practical attacks on symmetric cryptography



IETF Statement:

- **“The idea that symmetric cryptography will be practically affected by CRQCs is now seen as a misconception.** The “bits of security” concept does not work with algorithms that are not parallelizable and NIST is therefore transitioning to quantum-resistant security levels based on symmetric algorithms where level 1 is equivalent with AES-128, level 2 is SHA-256, etc. UK government assesses that “symmetric algorithms with at least 128-bit keys (such as AES) can continue to be used”. **While classical supercomputers might be able to brute force AES-128 around the year 2090, a huge cluster of one billion CRQCs (according to one estimate costing one billion USD each) would take a million years of uninterrupted calculation to find a single AES-128 key.** Algorithms with quadratic (n^2) speedup like Grover’s algorithm (which is proven to be optimal) will not provide any practical quantum advantage for breaking symmetric cryptography and likely not for any other problems.”

3GPP Statement:

- **“A very good summary of the impact of Cryptographically Relevant Quantum Computers (CRQCs) on symmetric cryptography was recently given in a statement by the Internet Engineering Task Force (IETF). The IETF statement refers to UK NCSC whitepaper that says symmetric algorithms with at least 128-bit keys (such as AES) can continue to be used. SA3 agrees with IETF’s analysis. Most other 128-bit algorithms such as SNOW 3G, ZUC, TUAKE, KMAC128, Ascon, etc. are likely to have similar quantum overhead for Grover’s algorithm which is known to be optimal.** Even if an algorithm has slightly lower quantum overhead than AES-128, SA3 believes the algorithm would still fulfill the requirement (comparable to key search on AES-128) for quantum resistance category 1”

(but 6G will use high-performance 256-bit algorithms)

Sam Jaques keynote at CHES 2024: “Qubits would cover the moon”

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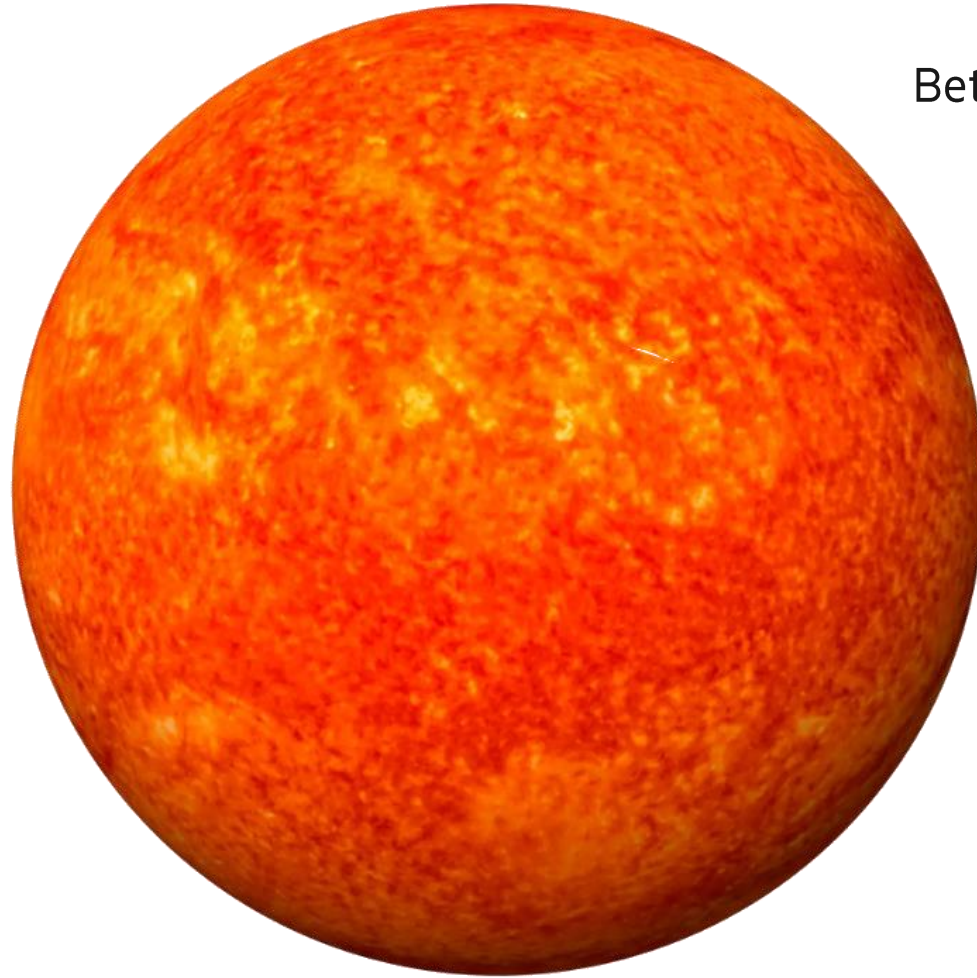


So, to break AES-128, you need qubits covering the surface area of the moon!
End of discussion?

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Our sun Sol



Betelgeuse

ECCG ACM 2025: Quantum threat to 191-bit symmetric keys...

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An atom-thick layer covering Betelgeuse would have the same volume as a 1 km thick layer covering Sweden...

Further reading



- IETF Statement on Quantum Safe Cryptographic Protocol Inventory
<https://datatracker.ietf.org/liaison/1942/>
- 3GPP Statement on PQC Migration
https://www.3gpp.org/ftp/tsg_sa/WG3_Security/TSGS3_118_Hyderabad/docs/S3-244307.zip
- Sam Jaques, "Quantum Attacks on AES "
<https://www.youtube.com/watch?v=eB4po9Br1YY&t=3227s>
- NIST NIST SP 800-57 Rev. 6 IPD
<https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-57pt1r6.ipd.pdf>
- ECCG Agreed Cryptographic Mechanisms - version 2
https://certification.enisa.europa.eu/document/download/a845662b-ae0-484e-9191-890c4cfa7aaq_en?filename=ECCG%20Agreed%20Cryptographic%20Mechanisms%20version%202.pdf
- Post-quantum cryptography in mobile networks
<https://www.ericsson.com/en/blog/2017/6/post-quantum-cryptography-in-mobile-networks>
- Migrating telecom to quantum-resistant cryptography on a global scale
<https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/migrating-telecom-to-quantum-resistant-cryptography>
- Lessons from a Decade of Post-Quantum Authentication
https://emanjon.github.io/Publications/Lessons_from_a_Decade_of_Post-Quantum_Authentication.pdf



